Adish Padalia

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Education

Rutgers University, New Brunswick, New Jersey **Aug 2024 - May 2026**
Master of Science in Computer Science GPA: 3.75/4

Sardar Patel Institute of Technology, University of Mumbai, India **Aug 2020 - May 2024**
Bachelor of Technology in Information Technology (Class Rank 4th) GPA: 3.87/4

Technical Skills

Programming Languages: Python, C/C++, Java, SQL, JavaScript, TypeScript
Frameworks: React.js, Next.js, Tailwind CSS, Flask, Node.js, Express.js, Django, Spring Boot, PHP, FastAPI, TensorFlow, PyTorch, Scikit-learn, LangChain, LLM Integration, RAG
Databases, Cloud & DevOps: MongoDB, PostgreSQL, SQLite, Firebase, AWS, Azure, GCP, Docker, Kubernetes, Prometheus, Grafana, Git, GitHub, Linux, Unix, CI/CD, Postman, Nginx

Experience

Graduate Research Assistant **June 2025 - Aug 2025**
Rutgers University New Jersey, United States

- Engineered a multimodal 3D object detection pipeline fusing LiDAR and camera data, improving 3D localization accuracy by 25% over 2D baseline models
- Developed a scalable auto-generation module for 3D bounding boxes (cuboids), reducing manual annotation and correction time by 40%
- Implemented 2D-to-3D geometric transformations with 98% reprojection accuracy, utilizing camera intrinsics and extrinsics to precisely map objects to real-world coordinates

Teaching Assistant **Sep 2024 - May 2026**
Rutgers University New Jersey, United States

- Mentored 200+ students teaching web development fundamentals, Excel, Java programming, database management, set theory, statistics and probability concepts
- Led weekly recitations, evaluated 75+ assignments and exams, and delivered lab and office-hour support to strengthen student understanding and academic performance

Research Intern **Jan 2023 - Jul 2023**
Indian Institute of Technology Bombay (in colloboration with AIIMs Delhi) Mumbai, India

- Innovated novel U-Net architectures integrating Discrete Wavelet Transform and Discrete Shearlet Transform for 3D Brain MRI segmentation, reducing model complexity to just 2k trainable parameters compared to millions in other state-of-the-art models
- Achieved 99% accuracy in 3D Brain MRI skull-stripping and multi-segmentation by benchmarking 120+ wavelet transforms
- Enhanced model interpretability by 25% using advanced gradient-weighting, developed a Shearlet transform library for neural networks and built a live 3D brain MRI segmentation interface

Machine Learning Intern **Sep 2022 - Dec 2022**
AI-DL Virtual Labs, Sardar Patel Institute of Technology Mumbai, India

- Launched a scalable full-stack virtual labs platform (Flask, React.js) enabling 400+ students to remotely practice AI/ML concepts
- Created 10 ML/DL experiments combining simulations, assessments, and real-world applications (Pose-Net, Backpropagation)

Projects

AI Agent for Licensing and Credentialing Automation
Flask, Next.js, React.js, AWS Textract, Gemini 2.5 Pro, PyMuPDF, AWS EC2/S3, RapidFuzz [Link]

- Automated a full-stack AI-enabled system that fills healthcare licensing forms directly from provider data, delivering over 85% accuracy and 70% faster processing by integrating AWS Textract OCR with Gemini 2.5 Pro
- Configured a modular Flask backend on AWS EC2/S3 cloud infrastructure to handle encrypted provider data and streamline the data ingestion pipeline, reducing system errors by 60%
- Reduced UI latency by 40% by leveraging a high-performance Next.js frontend to deliver a secure, responsive dashboard for quick uploads and PDF previews

FinSight AI – Multi-Agent LLM Platform for Personalized Financial Intelligence
LangGraph, LangChain, LangSmith, GCP Cloud Run, Docker, FastAPI, Next.js [Link]

- Architected and deployed a multi-tier, multi-agent financial intelligence platform on GCP Cloud Run, orchestrating 12 specialized AI agents using LangGraph and FastAPI to deliver personalized budget, investment, debt, and tax plans from a single user profile
- Implemented a multi-LLM routing system across 4 providers with per-agent model assignment and automatic fallbacks; integrated LangSmith observability, 6-layer safety guardrails, and automated GitHub Actions CI/CD with zero-downtime rollouts

Tech Debt Quantifier – AI-Powered Codebase Intelligence Platform
Python, FastAPI, LangChain, PostgreSQL, SQLAlchemy, Redis, Docker, Next.js, TypeScript, CI/CD [Link]

- Created a 3-agent AI pipeline (Crawler, Analyzer, Reporter) to quantify technical debt across 8 categories with a dollar-denominated cost model; structured 30+ REST API endpoints with PostgreSQL schema design and Redis async workers
- Incorporated a hybrid LLM mode (Meta Llama, OpenAI, Hugging Face) with formula-based fallback for reliability; integrated MCP servers for git history mining, Slack and Jira; deployed with Docker and CI/CD on a TypeScript and Next.js frontend

Publications

- **Shift-Invariant U-Net for Brain MRI Segmentation (ICACRS 2023, IEEE Xplore)**
- **Hybrid CNN and RNN for Twitter Sentiment Analysis (SmartCom 2024, Springer LNNS Vol. 947)**
- **Prediction and Management of Traffic Congestion in Urban Environments (ICCCNT 2024, IEEE Xplore)**

Achievements

- 3x Hackathon Winner: Columbia Get Hired Hack 2026: 1st of 50 teams; HackRU 2025: 1st of 450+ particpants and Top 5 of 200 teams at Barclays Data Stellar 2023 (team lead)